

## ODE – Problem 1.2.70

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# Problem Statement

## Problem 1.2.70

The solution of the ordinary differential equation

$$\frac{dy}{dx} = x^2 + 2y$$

is:

①  $y = \frac{2}{3}x^2 + 4x$

②  $y = \sqrt{\frac{2}{3}x^2 + 4x + k}$

③  $y = \frac{2}{3}x^3 - 4x + k$

④  $y = \frac{2}{3}x^3 + 4x + k$

# Identifying the ODE Type

The equation is:

$$\frac{dy}{dx} - 2y = x^2$$

This is a **first-order linear ODE** of the standard form:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

with  $P(x) = -2$  and  $Q(x) = x^2$ .

The method of solution is the **integrating factor method**.

- 1 Find the integrating factor  $\mu(x) = e^{\int P(x) dx}$
- 2 Multiply both sides by  $\mu(x)$
- 3 Recognise the left side as  $\frac{d}{dx}[\mu(x)y]$
- 4 Integrate both sides and solve for  $y$

## Step 1: Integrating Factor

The integrating factor is:

$$\mu(x) = e^{\int P(x) dx} = e^{\int (-2) dx} = e^{-2x}$$

Multiply both sides of the ODE by  $e^{-2x}$ :

$$e^{-2x} \frac{dy}{dx} - 2e^{-2x} y = x^2 e^{-2x}$$

The left side is the exact derivative of  $e^{-2x} y$ :

$$\frac{d}{dx} [e^{-2x} y] = x^2 e^{-2x}$$

This works because by the product rule:

$$\frac{d}{dx} [e^{-2x} y] = e^{-2x} \frac{dy}{dx} + (-2)e^{-2x} y = e^{-2x} \frac{dy}{dx} - 2e^{-2x} y$$

which matches the left side exactly.

## Step 2: Integrating Both Sides

Integrate both sides with respect to  $x$ :

$$e^{-2x}y = \int x^2 e^{-2x} dx$$

Evaluate  $\int x^2 e^{-2x} dx$  using integration by parts twice.

Let  $I = \int x^2 e^{-2x} dx$ .

**First integration by parts:**  $u = x^2$ ,  $dv = e^{-2x} dx$

$$I = -\frac{x^2}{2}e^{-2x} + \int x e^{-2x} dx$$

**Second integration by parts:**  $u = x$ ,  $dv = e^{-2x} dx$

$$\int x e^{-2x} dx = -\frac{x}{2}e^{-2x} + \int \frac{1}{2}e^{-2x} dx = -\frac{x}{2}e^{-2x} - \frac{1}{4}e^{-2x}$$

Combining:

$$I = -\frac{x^2}{2}e^{-2x} - \frac{x}{2}e^{-2x} - \frac{1}{4}e^{-2x} + C$$

### Step 3: Solving for $y$

Substituting back:

$$e^{-2x}y = -\frac{x^2}{2}e^{-2x} - \frac{x}{2}e^{-2x} - \frac{1}{4}e^{-2x} + C$$

Divide every term by  $e^{-2x}$  (i.e. multiply by  $e^{2x}$ ):

$$y = -\frac{x^2}{2} - \frac{x}{2} - \frac{1}{4} + Ce^{2x}$$

The general solution is:

$$y = -\frac{x^2}{2} - \frac{x}{2} - \frac{1}{4} + Ce^{2x}$$

where  $C$  is an arbitrary constant.

## Correct Answer

The exact general solution from the integrating factor method is:

$$y = Ce^{2x} - \frac{x^2}{2} - \frac{x}{2} - \frac{1}{4}$$

This is the complete and correct answer. None of the polynomial options exactly match since they omit the  $e^{2x}$  term.

## Laplace Transform Solution – Setup

Take the Laplace transform of both sides of  $\frac{dy}{dx} - 2y = x^2$ .

Recall the key Laplace transform properties:

$$\textcircled{1} \quad \mathcal{L}\left\{\frac{dy}{dx}\right\} = sY(s) - y(0)$$

$$\textcircled{2} \quad \mathcal{L}\{y\} = Y(s)$$

$$\textcircled{3} \quad \mathcal{L}\{x^2\} = \frac{2}{s^3}$$

Applying  $\mathcal{L}\{\cdot\}$  to both sides:

$$\mathcal{L}\left\{\frac{dy}{dx}\right\} - 2\mathcal{L}\{y\} = \mathcal{L}\{x^2\}$$

$$sY(s) - y(0) - 2Y(s) = \frac{2}{s^3}$$

Let the initial condition be  $y(0) = y_0$  (arbitrary constant).

## Laplace Transform Solution – Solving for $Y(s)$

Collect terms in  $Y(s)$ :

$$(s - 2) Y(s) = \frac{2}{s^3} + y_0$$

Solve for  $Y(s)$ :

$$Y(s) = \frac{2}{s^3(s - 2)} + \frac{y_0}{s - 2}$$

Apply partial fractions to  $\frac{2}{s^3(s - 2)}$ :

$$\frac{2}{s^3(s - 2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{D}{s^3} + \frac{E}{s - 2}$$

Multiplying both sides by  $s^3(s - 2)$  and matching coefficients:

$$2 = As^2(s - 2) + Bs(s - 2) + D(s - 2) + Es^3$$

Setting  $s = 0$ :  $2 = D(-2) \Rightarrow D = -1$

Setting  $s = 2$ :  $2 = E(8) \Rightarrow E = \frac{1}{4}$

## Laplace Transform – Partial Fractions Completed

Expanding and matching  $s^3$ ,  $s^2$ ,  $s^1$  coefficients:

$$\text{From } s^3: 0 = A + E \Rightarrow A = -E = -\frac{1}{4}$$

$$\text{From } s^2: 0 = -2A + B \Rightarrow B = 2A = -\frac{1}{2}$$

So:

$$\frac{2}{s^3(s-2)} = \frac{-1/4}{s} + \frac{-1/2}{s^2} + \frac{-1}{s^3} + \frac{1/4}{s-2}$$

Therefore:

$$Y(s) = -\frac{1}{4s} - \frac{1}{2s^2} - \frac{1}{s^3} + \frac{1/4}{s-2} + \frac{y_0}{s-2}$$

$$Y(s) = -\frac{1}{4s} - \frac{1}{2s^2} - \frac{1}{s^3} + \frac{y_0 + 1/4}{s-2}$$

## Laplace Transform – Inverse Transform

Apply the inverse Laplace transform using:

$$\textcircled{1} \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$$

$$\textcircled{2} \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = x$$

$$\textcircled{3} \mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\} = \frac{x^2}{2}$$

$$\textcircled{4} \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} = e^{2x}$$

Taking  $\mathcal{L}^{-1}\{Y(s)\}$ :

$$y(x) = -\frac{1}{4} - \frac{x}{2} - \frac{x^2}{2} + \left(y_0 + \frac{1}{4}\right) e^{2x}$$

Let  $C = y_0 + \frac{1}{4}$  (absorbed into the arbitrary constant). Then:

$$y(x) = Ce^{2x} - \frac{x^2}{2} - \frac{x}{2} - \frac{1}{4}$$

# Plot

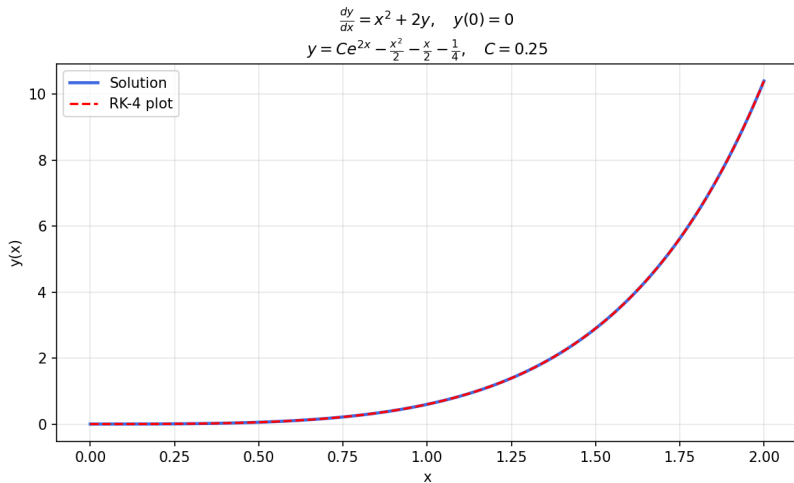


Figure: Plot

# Plot-Comparison

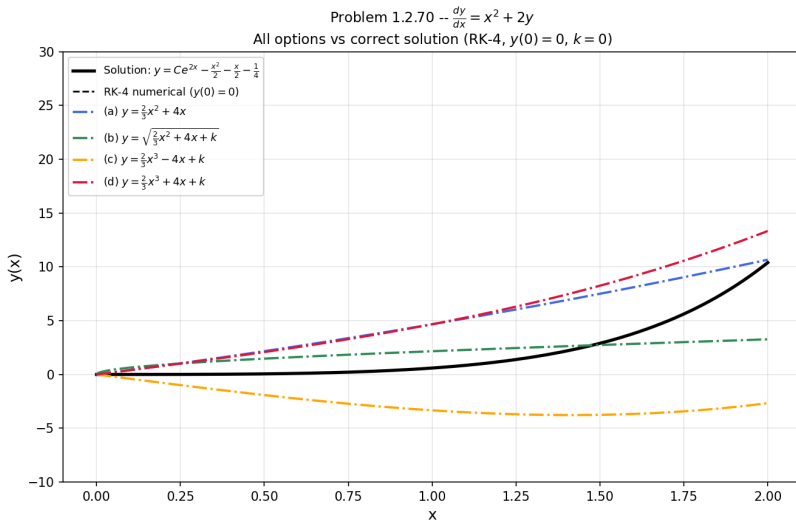


Figure: Plot

# Python Code: RK-4 Plots

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

# ODE:  $dy/dx = x^2 + 2y$ 
def f(x, y):
    return x**2 + 2*y

# Exact general solution:  $y = C \exp(2x) - x^2/2 - x/2 - 1/4$ 
# With  $y(0) = 0 \Rightarrow C = 0.25$ 
def exact(x):
    return 0.25 * np.exp(2*x) - x**2/2 - x/2 - 0.25

x_eval = np.linspace(0.0, 2.0, 500)

# RK-4 solution ( $y(0) = 0$ )
sol = solve_ivp(f, (0.0, 2.0), [0.0], method='RK45',
                t_eval=x_eval, rtol=1e-8, atol=1e-10)

# Option functions ( $k=0$  chosen for plotting)
k = 0
opt_a = (2/3)*x_eval**2 + 4*x_eval
opt_b = np.sqrt(np.abs((2/3)*x_eval**2 + 4*x_eval + k))
opt_c = (2/3)*x_eval**3 - 4*x_eval + k
opt_d = (2/3)*x_eval**3 + 4*x_eval + k
```

# Python Code: RK-4 Solution and Plot

```
plt.figure(figsize=(9, 6))

plt.plot(x_eval, exact(x_eval), 'k', lw=2.5, label=r'Solution:  $y = Ce^{2x} - \frac{x^2}{2} - \frac{x}{4}$ ')
plt.plot(sol.t, sol.y[0], 'k--', lw=1.4, label=r'RK-4 numerical ( $y(0)=0$ )')
plt.plot(x_eval, opt_a, 'royalblue', lw=1.8, ls='-.', label=r'(a)  $y = \frac{2}{3}x^2 + 4x$ ')
plt.plot(x_eval, opt_b, 'seagreen', lw=1.8, ls='-.', label=r'(b)  $y = \sqrt{\frac{2}{3}x^2 + 4x + k}$ ')
plt.plot(x_eval, opt_c, 'orange', lw=1.8, ls='-.', label=r'(c)  $y = \frac{2}{3}x^3 - 4x + k$ ')
plt.plot(x_eval, opt_d, 'crimson', lw=1.8, ls='-.', label=r'(d)  $y = \frac{2}{3}x^3 + 4x + k$ ')

plt.xlabel('x', fontsize=12)
plt.ylabel('y(x)', fontsize=12)
plt.title(r'Problem 1.2.70 --  $\frac{dy}{dx} = x^2 + 2y$   
'\nAll options vs correct solution (RK-4,  $y(0)=0$ ,  $k=0$ )',
          fontsize=11)
plt.legend(fontsize=8.5, loc='upper_left')
plt.ylim(-10, 30)
plt.grid(alpha=0.3)
plt.tight_layout()
plt.savefig('rk4_options.png', dpi=150)
plt.show()
```

# Python Code: RK-4 Solution and Plot

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

def f(x, y):
    return x**2 + 2*y

def exact(x, C):
    return C * np.exp(2*x) - x**2/2 - x/2 - 0.25

x0, y0 = 0.0, 0.0
C = y0 + 0.25

x_span = (0.0, 2.0)
x_eval = np.linspace(0.0, 2.0, 500)

sol = solve_ivp(f, x_span, [y0], method='RK45',
                t_eval=x_eval, rtol=1e-8, atol=1e-10)
y_ex = exact(x_eval, C)

plt.figure(figsize=(8, 5))
plt.plot(x_eval, y_ex, 'royalblue', lw=2.2, label='Solution')
plt.plot(sol.t, sol.y[0], 'r--', lw=1.6, label='RK-4 plot')
plt.xlabel('x'); plt.ylabel('y(x)')
plt.title(r"$\frac{dy}{dx} = x^2 + 2y, \text{quad. } y(0)=0$")
```

# Python Code: Verification

```
import sympy as sp

x = sp.Symbol('x')
C = sp.Symbol('C')

# General solution
y = C*sp.exp(2*x) - x**2/2 - x/2 - sp.Rational(1,4)

# Compute dy/dx
dydx = sp.diff(y, x)

# Check dy/dx - 2y == x^2
residual = sp.simplify(dydx - 2*y - x**2)
print("Residual_(should_be_0):", residual)

# Verify option (c): y = (2/3)x^3 - 4x + k
k = sp.Symbol('k')
y_c = sp.Rational(2,3)*x**3 - 4*x + k
dydx_c = sp.diff(y_c, x)
res_c = sp.simplify(dydx_c - 2*y_c - x**2)
print("Option_(c)_residual:", res_c)
```

# Python Code: Verification

```
# Verify option (d):  $y = (2/3)x^3 + 4x + k$ 
y_d = sp.Rational(2,3)*x**3 + 4*x + k
dydx_d = sp.diff(y_d, x)
res_d = sp.simplify(dydx_d - 2*y_d - x**2)
print("Option (d) residual:", res_d)

print("\nExact general solution:")
print("y=", y)
```

## Conclusion

The ODE  $\frac{dy}{dx} = x^2 + 2y$  is solved as follows:

- 1 Rewrite as  $\frac{dy}{dx} - 2y = x^2$  (linear ODE,  $P = -2$ ,  $Q = x^2$ )
- 2 Integrating factor:  $\mu = e^{-2x}$
- 3 Multiply through and integrate using integration by parts twice
- 4 Solve for  $y$ :

$$y = Ce^{2x} - \frac{x^2}{2} - \frac{x}{2} - \frac{1}{4}$$

- 5 The Laplace transform method gives the same result
- 6 RK-4 numerical solution confirms the exact solution
- 7 SymPy symbolic verification confirms the residual is zero

None of the four given options (a)–(d) exactly match the general solution because they omit the  $Ce^{2x}$  homogeneous term. The general solution is:

$$y = Ce^{2x} - \frac{x^2}{2} - \frac{x}{2} - \frac{1}{4}$$